Aadish Bane

AI/ML & Software Engineer — LLM systems, agentic AI, speech processing
aadishbane23022005@gmail.com | linkedin.com | github.com

Education

B.Tech, Electronics & Communication Engineering — Gujarat, India
Sardar Vallabhbhai National Institute of Technology (SVNIT), Surat — *2023 – 2027*
- Relevant coursework: Machine Learning, Generative AI, Responsible AI, Speech Processing, Data Structures & Algorithms, Probability & Statistics.

Experience

AI/ML Intern — Remote
Deevelo X — *May 2025 – Jul 2025*
- Built an LLM-powered document generation application on Mixtral-8x7B with STT/TTS integration, category/urgency classification and automated formal PDF output, supporting 1,000+ generations.
- Designed prompt templates and a classification layer that tailored output across real-world scenarios, improving relevance over a single-prompt baseline.
- Built supervised ML models on 10,000+ sample datasets, improving performance through feature engineering; evaluated with cross-validation and GridSearchCV.

AI Intern — Abu Dhabi, UAE
Abu Dhabi National Oil Company (ADNOC) — *Dec 2025 – Jan 2026*
- Designed PLC ladder logic and wrote PLC code for industrial control workflows on live SCADA-integrated systems.
- Mapped integration pathways between AI/ML models and PLC-based control systems, assessing feasibility of AI-driven automation within existing SCADA/DCS infrastructure.
- Assessed AI adoption opportunities across SCADA, PLC and DCS systems; presented findings and recommendations on efficiency gains from AI-based monitoring.

Projects

Intelligent LLM Routing & Cost Optimization Platform
Python, FastAPI, Streamlit, SQLite, scikit-learn, Docker
- Built an 11-phase routing service that predicts cost across 12 models from 6 providers before every call and routes to the cheapest model meeting a quality bar.
- Added a TF-IDF retrieval layer for context compression (60%+ token reduction) and a multi-objective optimization engine with automated quality-verification/escalation.
- Demonstrated a simulated 70.8% cost reduction versus always calling the most expensive model; exposed as a FastAPI REST service with a Streamlit analytics dashboard and audit logging.

Speech Emotion Recognition — MSA-ECN Architecture
Python, PyTorch, Librosa
- Hybrid Multi-Scale CNN + Conformer on the RAVDESS dataset, with MFCC, Mel Spectrogram and Chroma features; 85% classification accuracy.

AdaptiQuant — Regime-Aware Multi-Agent Trading Backtester
Python, pandas, NumPy, scikit-learn, SHAP
- Event-driven engine simulating 10+ years of market data (2,300+ bars) with commission, slippage and risk controls.
- Multi-agent framework classifying 4 market regimes from 30+ technical indicators, with a meta-agent scoring strategies on return, Sharpe, win-rate and drawdown.

Responsible AI Governance Framework
Python, Groq API, Streamlit, SHAP, Fairlearn, Isolation Forest
- Fairness checks, bias detection, policy enforcement and RBAC with a Streamlit dashboard for SHAP explainability, audit logging, anomaly detection and compliance monitoring.

Skills

Python, FastAPI, PyTorch, scikit-learn, pandas / NumPy, LLMs & GenAI (Mixtral, agentic workflows, prompt engineering, LLM routing/eval), Speech & audio ML (Librosa, CNN, Conformer), Streamlit, Docker, SQLite, Git / GitHub, SHAP, Fairlearn